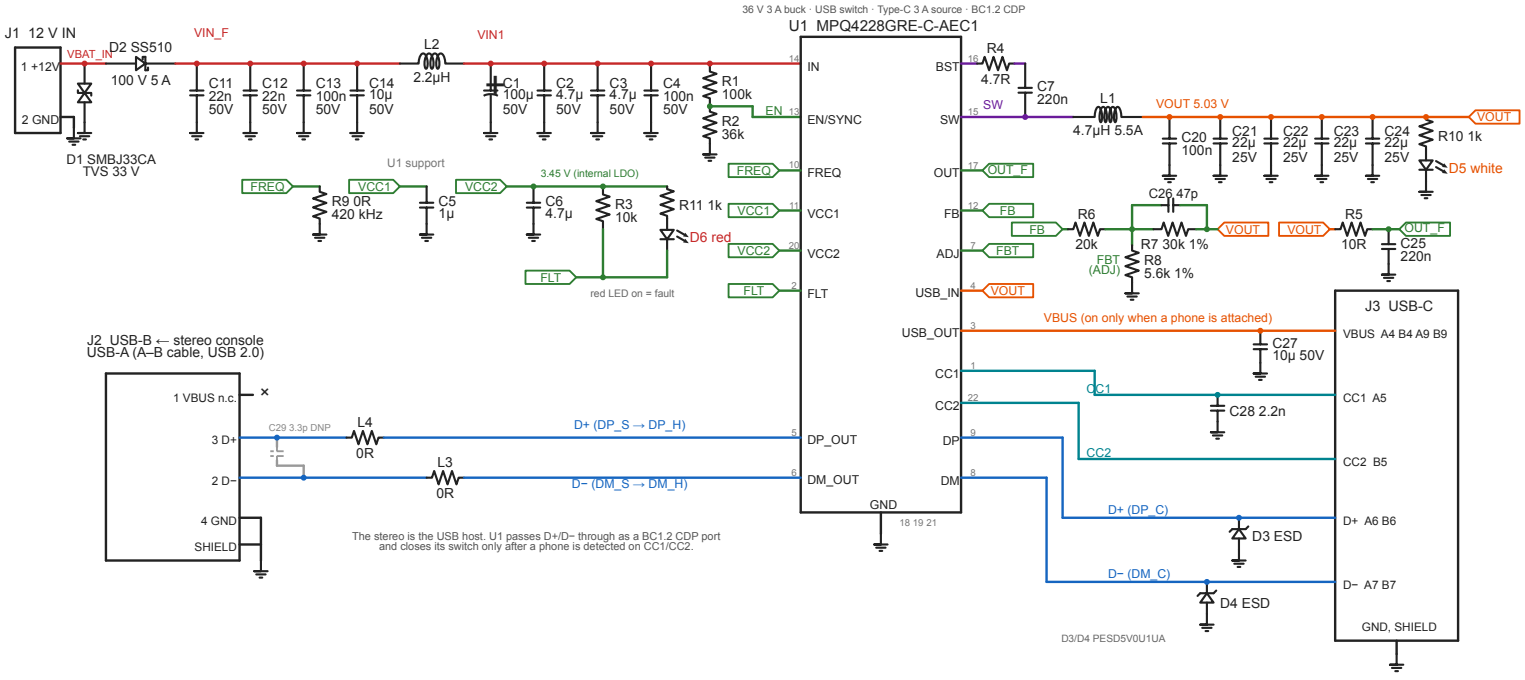


IN THE CAR — CONNECTIONS

J1 +12V ← fuse tap on an ACC (switched) fuse, 5 A fuse. Swift: fuse 9 ACC-2 or 15 ACC.
J1 GND → body ground at the stereo's ground point (solid, short).
J2 USB-B ← stereo console USB-A socket, USB 2.0 A–B cable (stereo 5 V unused).
J3 USB-C → full-featured USB-C extension (CC1/CC2 separate) → phone.
LEDs: WHITE = 5 V output on. RED = fault (overload, short, over-voltage, over-temperature); clears itself when the fault goes.
VBUS on J3 reads 0 V until a phone is attached (normal Type-C behaviour).

RATINGS

Input: 12 V car supply; on above ≈6.1 V, off below ≈5.5 V
Max input 36 V continuous (U1 40 V abs.)
Protection: D1 TVS 33 V bidirectional, D2 reverse-polarity block, 5 A fuse
Output: 5.03 V, 3 A (15 W), Type-C Rp 3 A, BC1.2 CDP, cable-drop comp. +0.18 V at 3 A
VOUT = 0.792 V × (1 + R7/R8) + R7 × 2.05 µA/A × I
Switching 420 kHz, spread spectrum; no USB PD
Wire colours: red 12 V, purple SW, orange 5 V, blue USB data, teal CC, green control

SWIFT USB-C 3A

Car 12 V → USB-C 5 V 3 A charger with stereo USB data pass-through (iPod protocol)

Vehicle	2011 Suzuki Swift
Board	swift-usbc-3a, 4-layer, 52 × 40 mm
Rev	v1
Date	07/10/2026
Sheet	1 of 2 (schematic)
Source	KiCad: ~/kicad/swift-usbc-3a

SWIFT USB-C 3A — PARTS LIST (as assembled by JLCPCB, 07/10/2026)

C29 (3.3 pF) not fitted. L3/L4 fitted as 0 Ω links (18 nH optional). J1/J2 hand-soldered by JLCPCB. Parts from LCSC (lcsc.com/product-detail/<LCSC>.html).

Ref	Value	Package	LCSC	Role
C1	100µF 50V	Ø8 × 10.5 mm SMD	C19184158	input bulk
C2, C3	4.7µF 50V	0805	C20416422	input, at U1
C4, C13, C20	100nF 50V	0603	C14663	decoupling
C5	1µF	0603	C15849	VCC1
C6	4.7µF	0603	C19666	VCC2
C7, C25	220nF	0603	C21120	bootstrap / OUT filter
C11, C12	22nF 50V	0603	C21122	input filter
C14, C27	10µF 50V	1206	C13585	input filter / VBUS
C21, C22, C23, C24	22µF 25V	1206	C12891	output
C26	47pF C0G	0603	C1671	feed-forward
C28	2.2nF	0603	C1604	CC1 filter
D1	SMBJ33CA	SMB	C19077587	TVS 33 V bidirectional
D2	SS510	SMA	C18199171	reverse block
D3, D4	PESD5V0U1UA	SOD-323	C19829579	ESD on D+/D−
D5	White LED	0603	C2290	power LED
D6	FAULT red	0603	C2286	fault LED
J1	12V IN	5.08 mm 2-pin, through-hole	C7509570	12 V input
J2	USB-B from stereo	USB-B, through-hole	C19944605	from stereo
J3	USB-C to phone	USB-C 16-pin SMD	C165948	to phone
L1	4.7µH 3.6A	6 × 6 mm	C6808109	buck inductor
L2	2.2µH 4.7A	3 × 3 mm	C7423318	input filter
L3, L4, R9	0R	0603	C21189	0 Ω links (D+/D−, FREQ)
R1	100k	0603	C25803	EN divider
R2	36k 1%	0603	C23147	EN divider
R3	10k	0603	C25804	FLT pull-up
R4	4.7R	0603	C23164	bootstrap
R5	10R	0603	C22859	OUT filter
R6	20k	0603	C4184	FB (RT)
R7	30k 1%	0603	C22984	VOUT set
R8	5.6k 1%	0603	C23189	VOUT set
R10, R11	1k	0603	C21190	LED resistors
U1	MPQ4228-C	QFN-22 4 × 4 mm	C17638787	buck + USB switch